

```

> with(DEtools) :
> ecdif := diff(y(x), x) = 2 · x · (1 + ( y(x) )^2) :
> dsolve(ecdif, y(x));

```

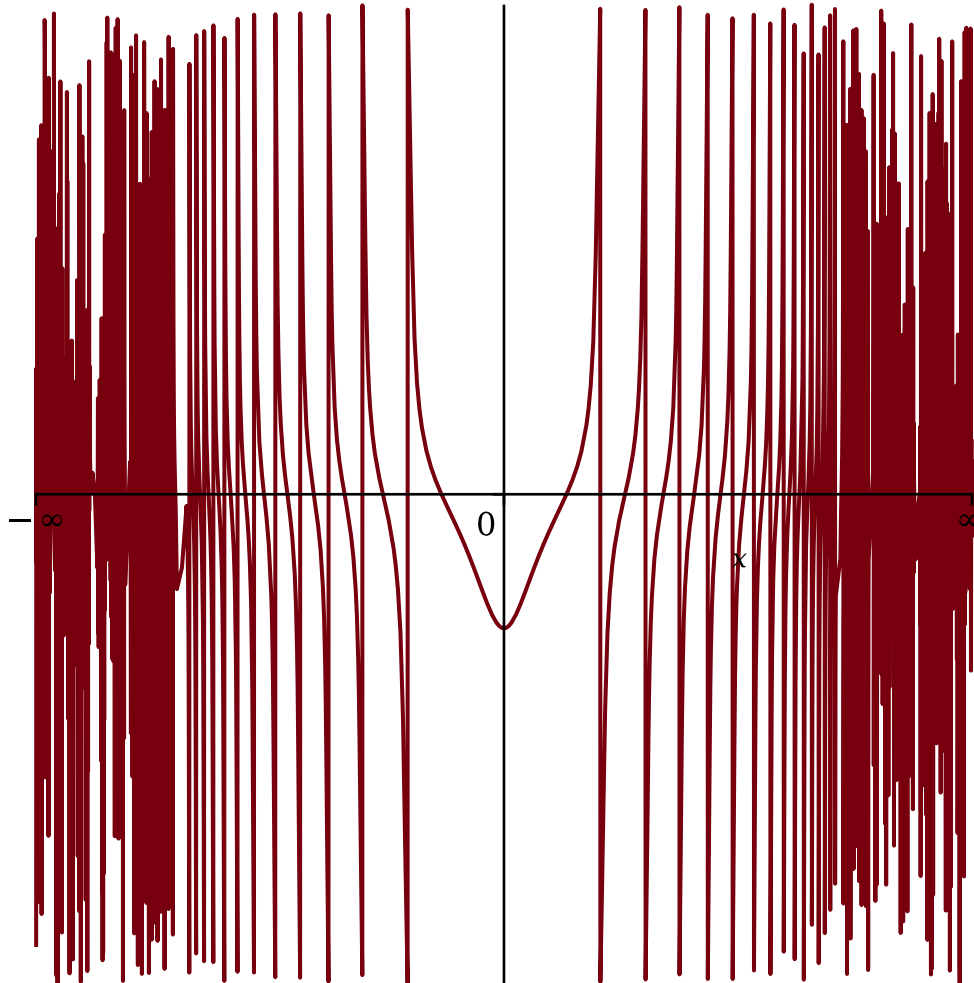
$$y(x) = \tan(x^2 + 2 c_1)$$

(1)

```

> y1 := (x, c) → tan(x^2 + 2 · c) :
> with(plots) :
> plot(y1(x, 1), x = -infinity .. infinity)

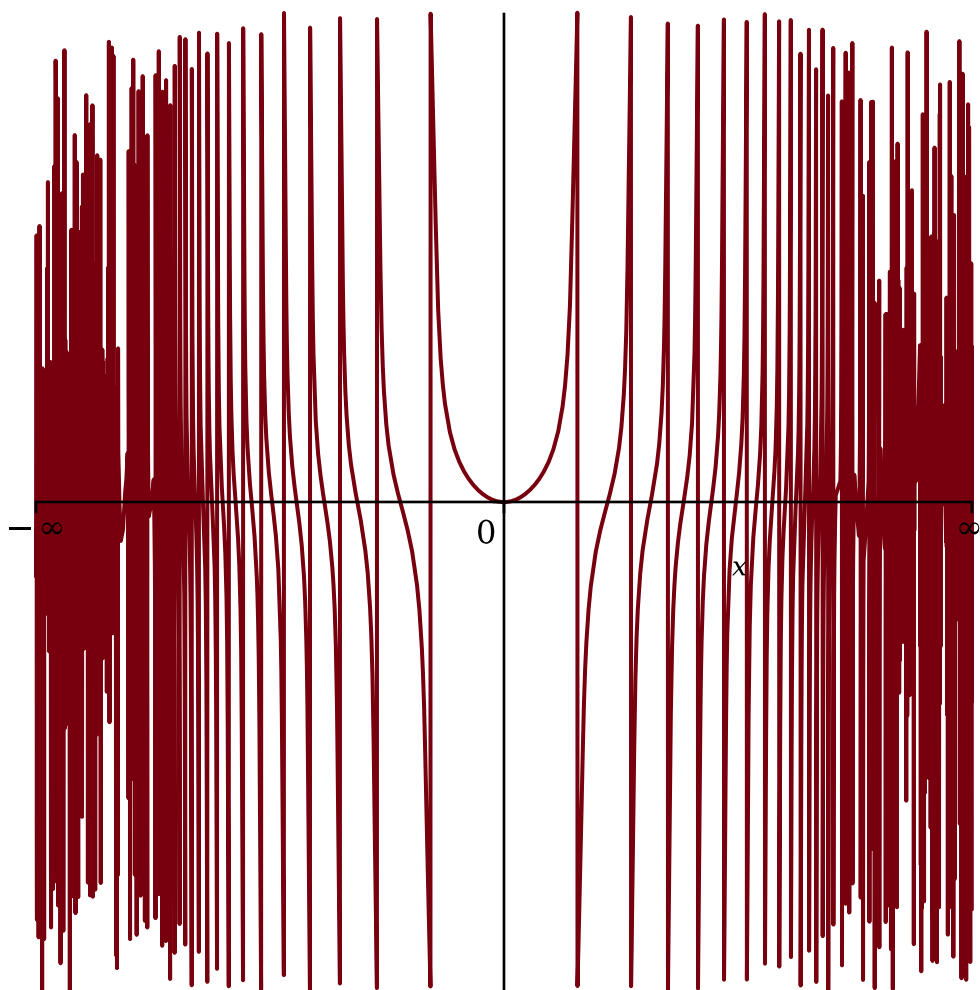
```



```

> y1 := (x, c) → tan(x^2 + 2 · c) :
> with(plots) :
> plot(y1(x, 314/100), x = -infinity .. infinity)

```



```
> with(DEtools):
```

```
> ecdif := (x^2 - 1) · diff(y(x), x) + 2 · x · (y(x))^2 = 0:
```

```
> dsolve(ecdif, y(x));
```

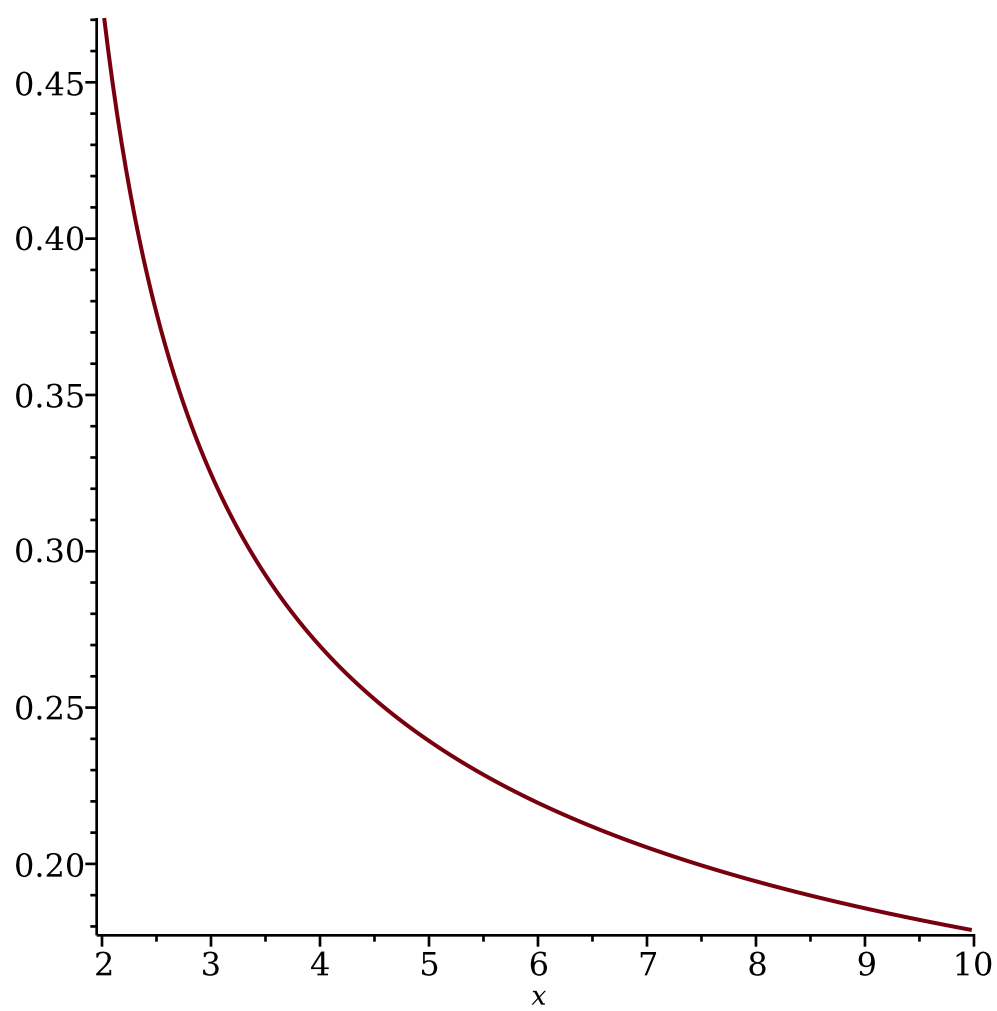
$$y(x) = \frac{1}{\ln(x-1) + \ln(x+1) + c_1}$$

(2)

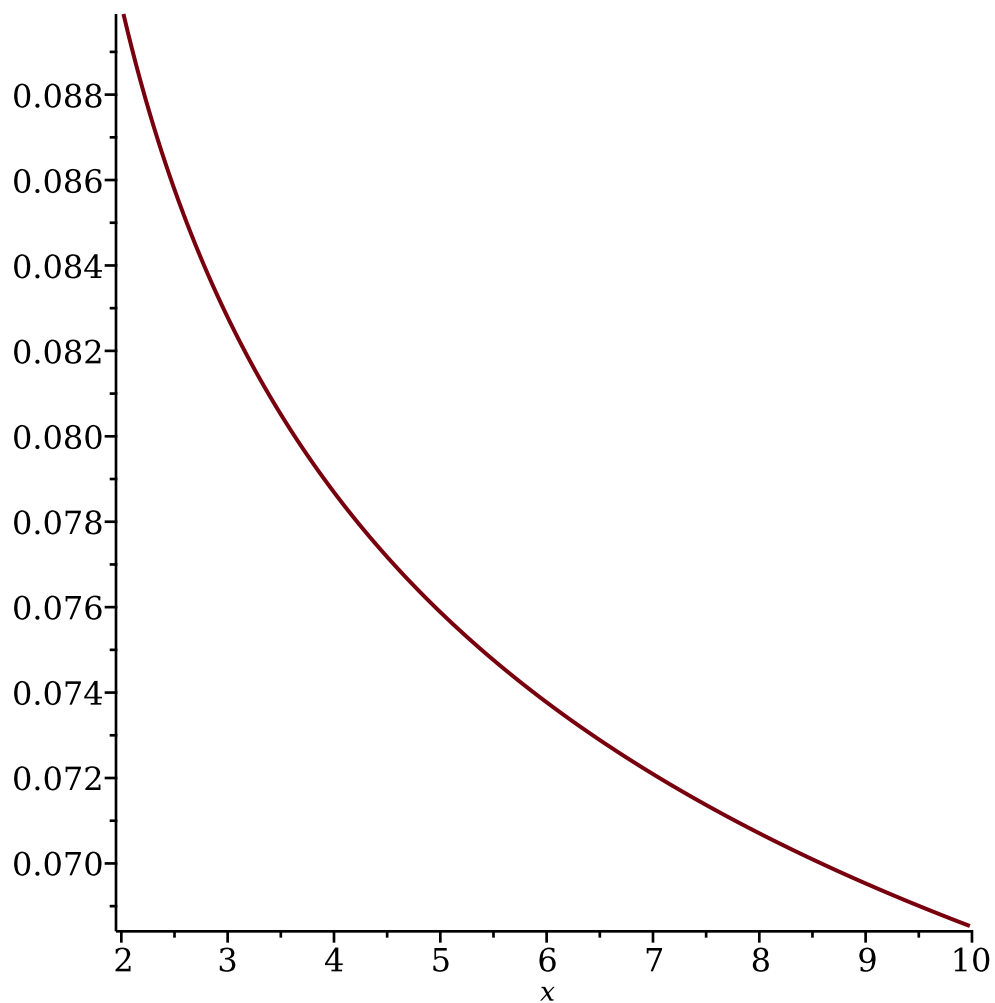
```
> y1 := (x, c) → 1 / (ln(x-1) + ln(x+1) + c):
```

```
> with(plots):
```

```
> plot(y1(x, 1), x = 2..10)
```



```
> plot(y1(x, 10), x = 2..10)
```



```
> with(DEtools):
```

```
> ecdif:= 2 ·x2·diff(y(x), x) = x2 + (y(x))2:
```

```
> dsolve(ecdif, y(x))
```

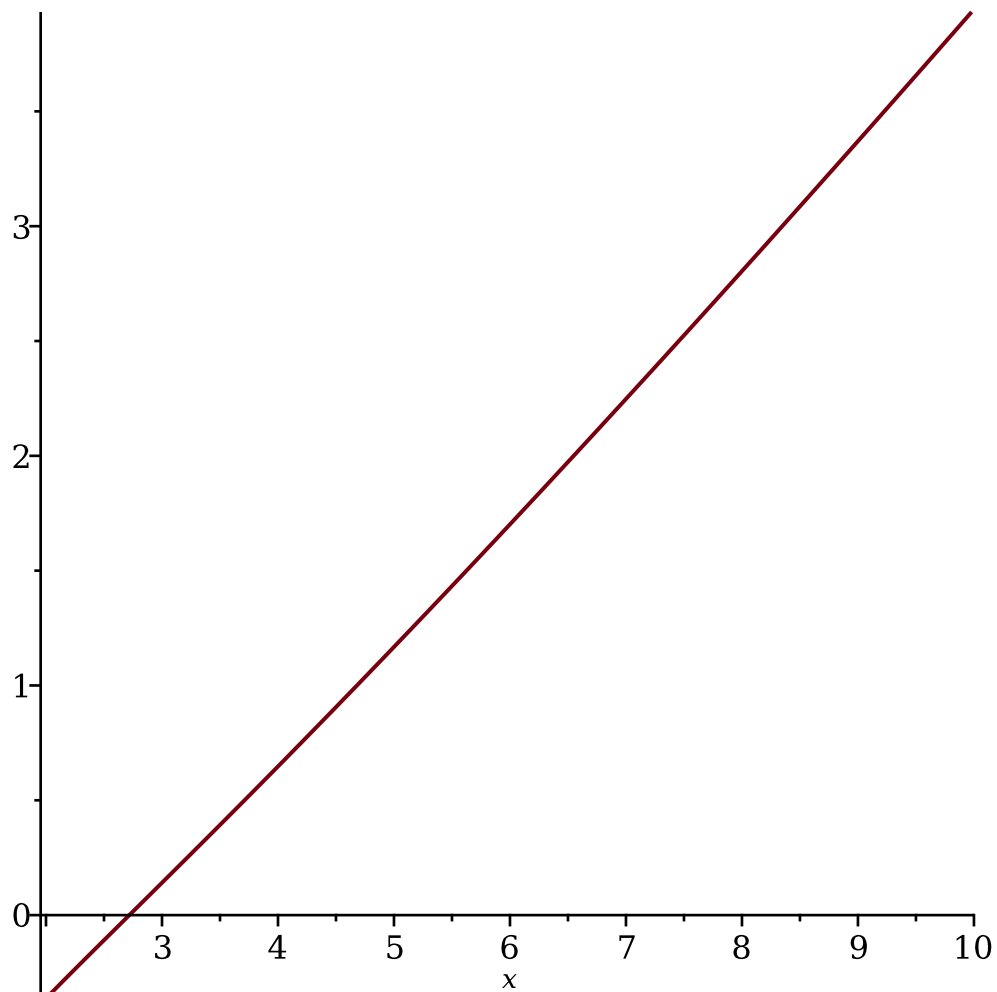
$$y(x) = \frac{x (\ln(x) + c_1 - 2)}{\ln(x) + c_1}$$

**(3)**

```
> y1 := (x, c)→  $\frac{(x \cdot (\ln(x) + c - 2))}{\ln(x) + c}$  :
```

```
> with(plots):
```

```
> plot(y1(x, 1), x = 2..10)
```



```
> with(DEtools):
```

```
> ecdif := diff(y(x), x) = - ( x / y(x) ) :
```

```
> dsolve(ecdif, y(x));
```

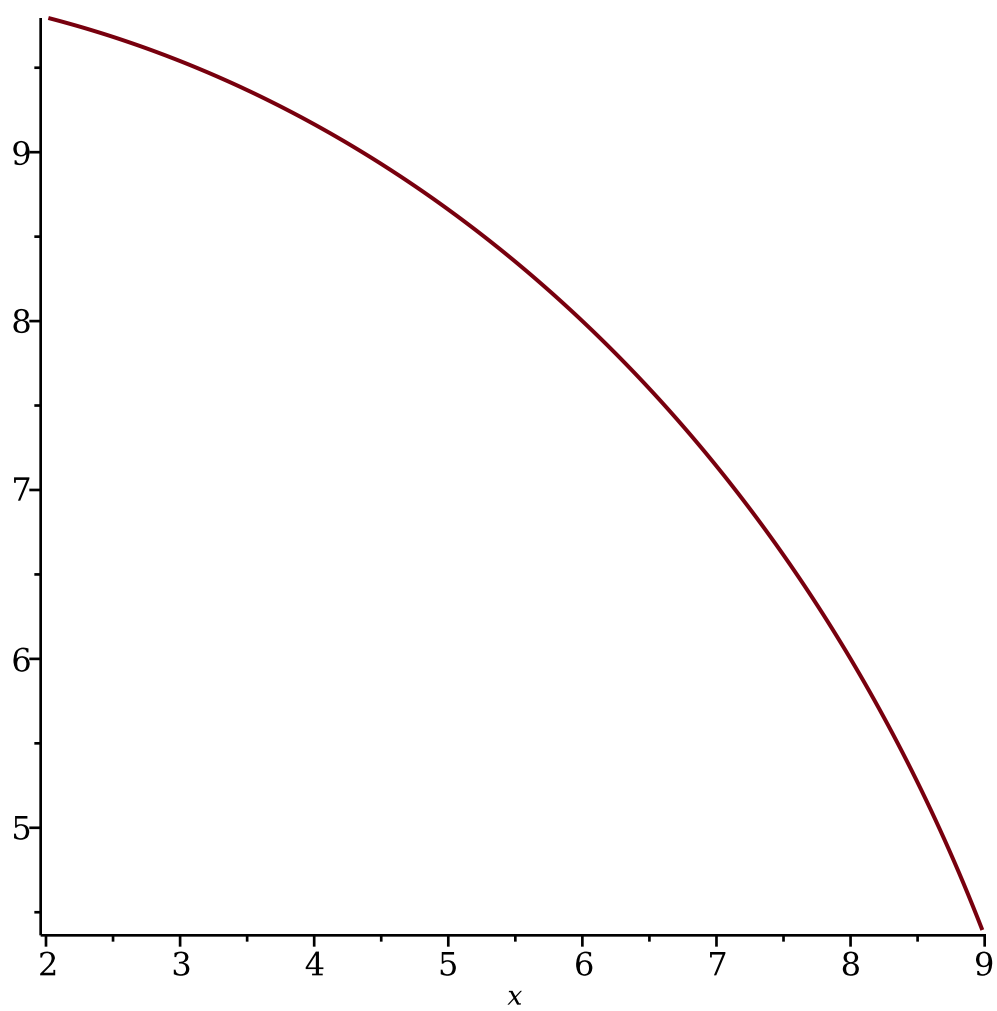
$$y(x) = \sqrt{-x^2 + c_1}, y(x) = -\sqrt{-x^2 + c_1}$$

**(4)**

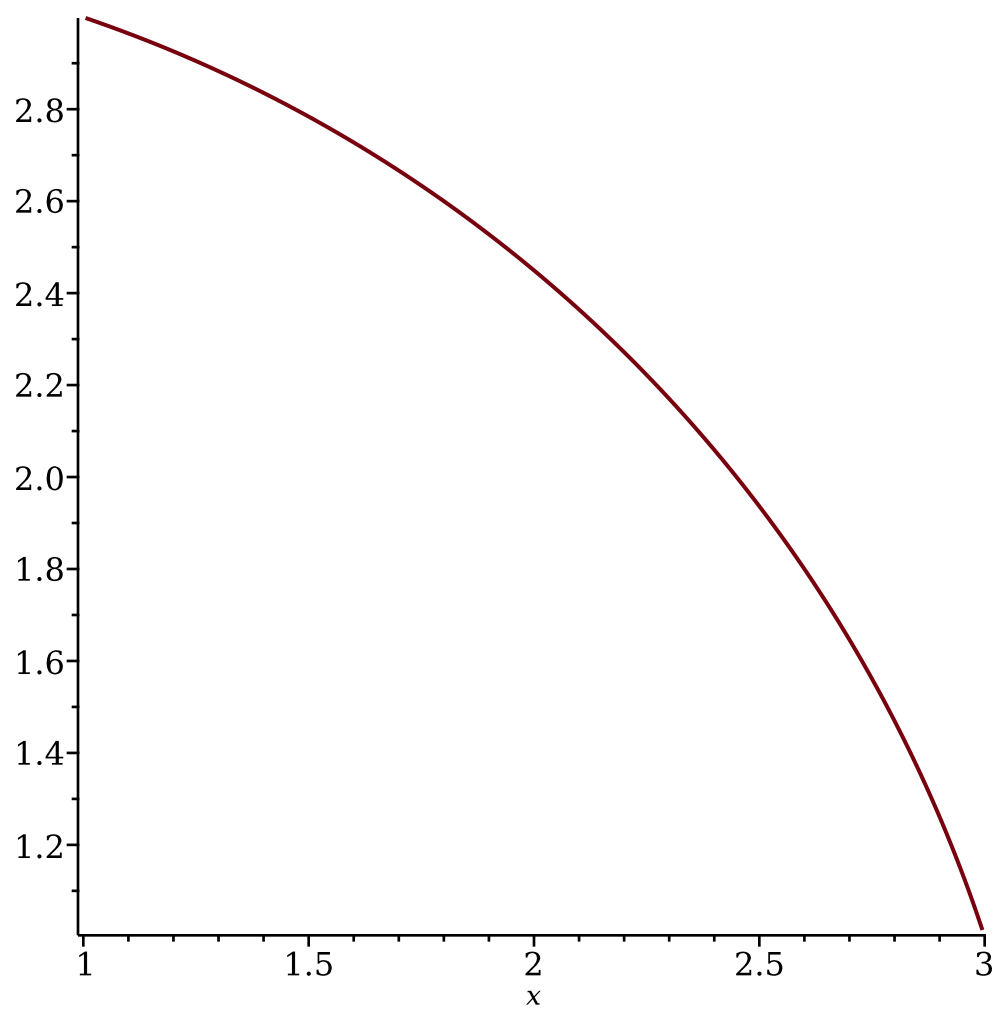
```
> y1 := (x, c) -> sqrt(-x^2 + c) :
```

```
> with(plots) :
```

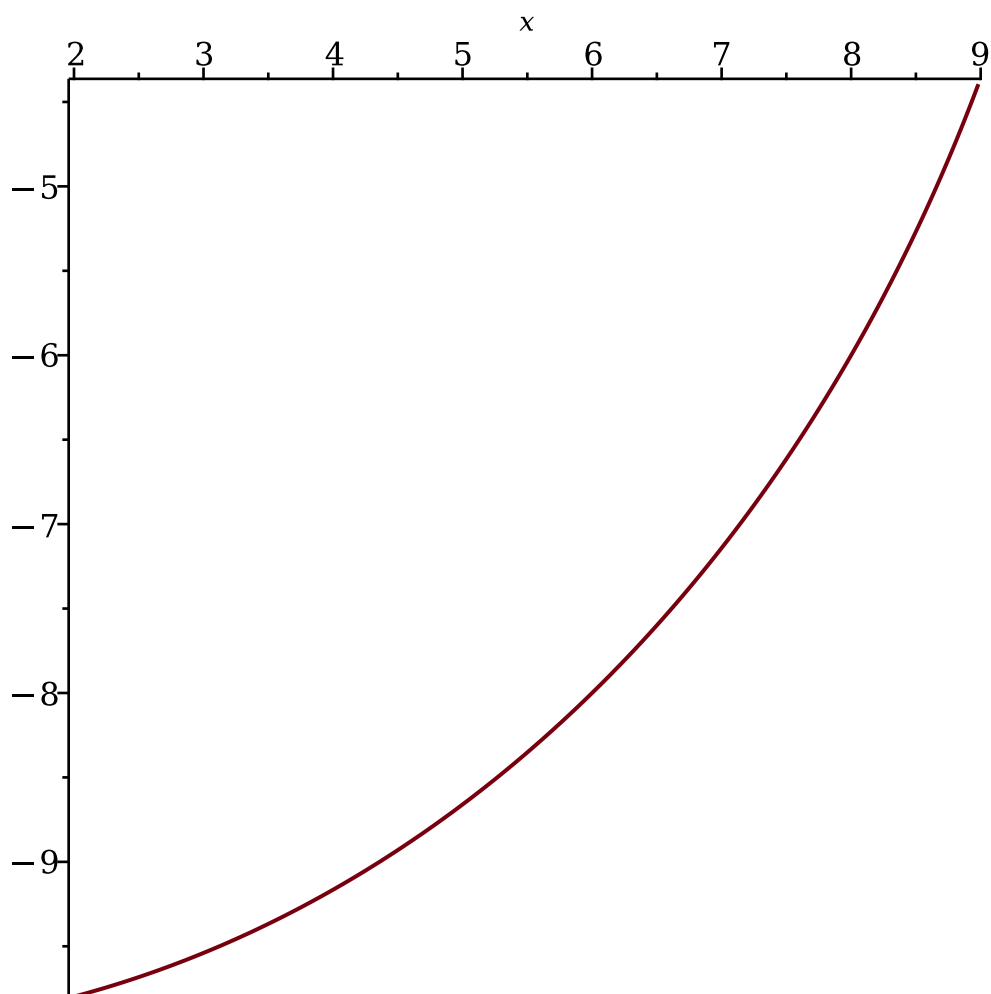
```
> plot(y1(x, 100), x = 2..9)
```



```
> plot(y1(x, 10), x = 1..3)
```

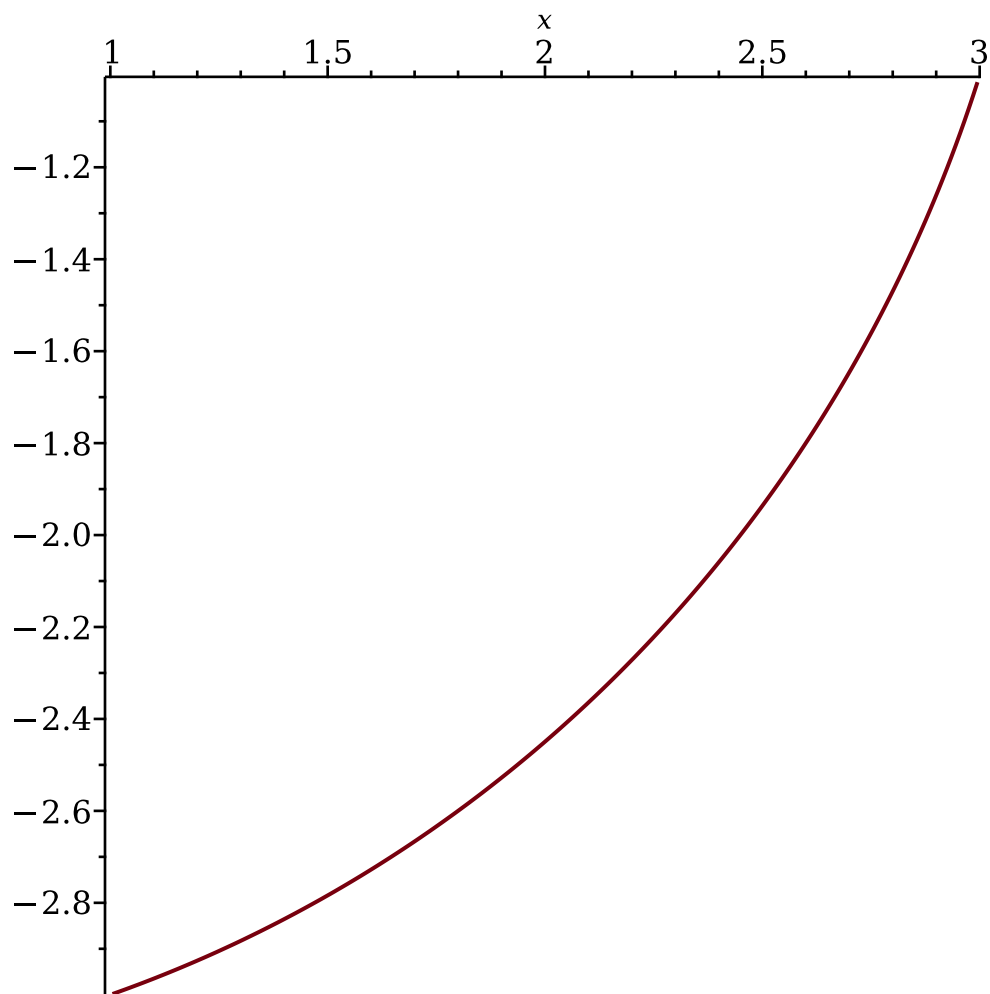


```
>  
y2 := (x, c) -> -sqrt(-x^2 + c) :  
with(plots) :  
> plot(y2(x, 100), x = 2..9)
```



```
>  
y2 := (x, c) -> -sqrt(-x^2 + c) :  
with(plots) :  
> plot(y2(x, 10), x = 1..9)
```

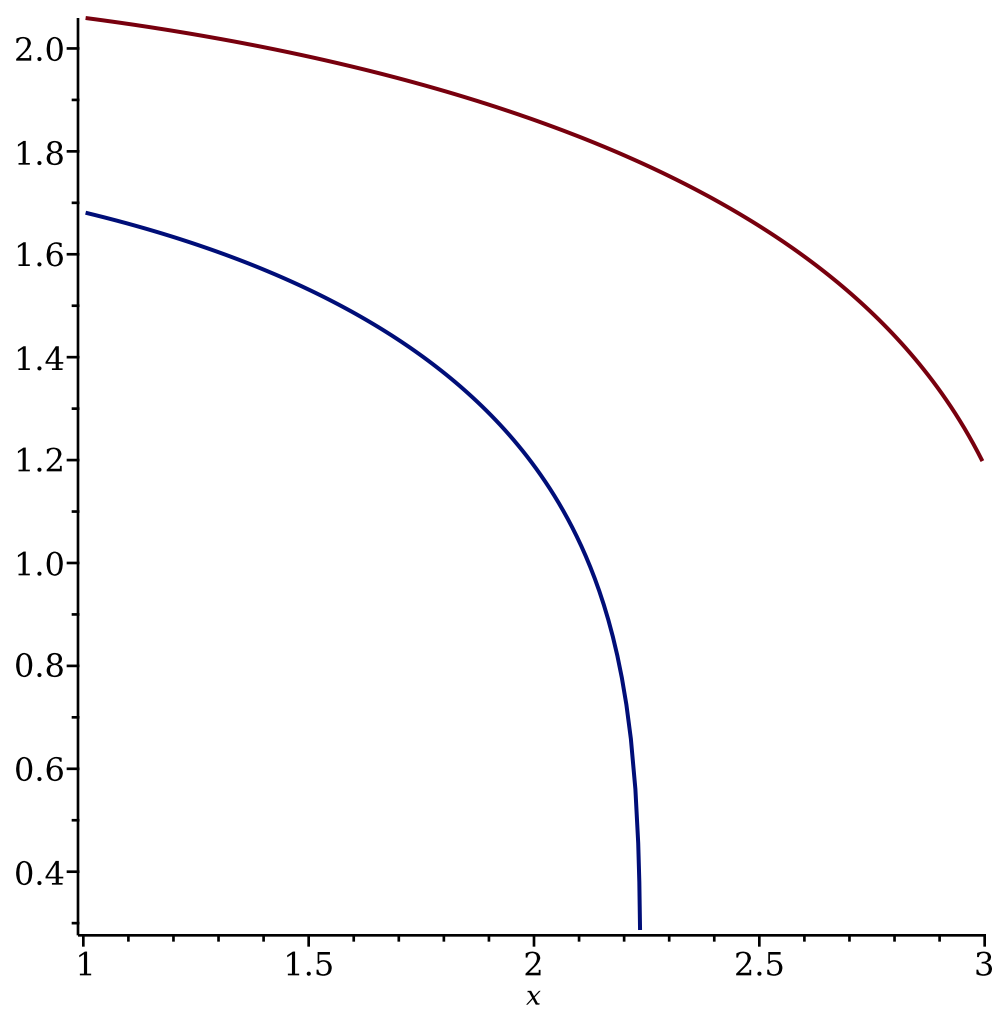




```

=>
=> with(DEtools) :
=> ecdif:=diff(y(x), x) = - ( x / (y(x))^3 ) :
=> dsolve(ecdif, y(x));
y(x) = (-2 x^2 + c_1)^(1/4), y(x) = -(-2 x^2 + c_1)^(1/4), y(x) = -I (-2 x^2 + c_1)^(1/4), y(x) (5)
      = I (-2 x^2 + c_1)^(1/4)
=>
=> y1 := (x, c) -> (-2*x^2 + c)^(1/4) :
=> with(plots) :
=> plot([y1(x, 20), y1(x, 10)], x = 1..3);

```

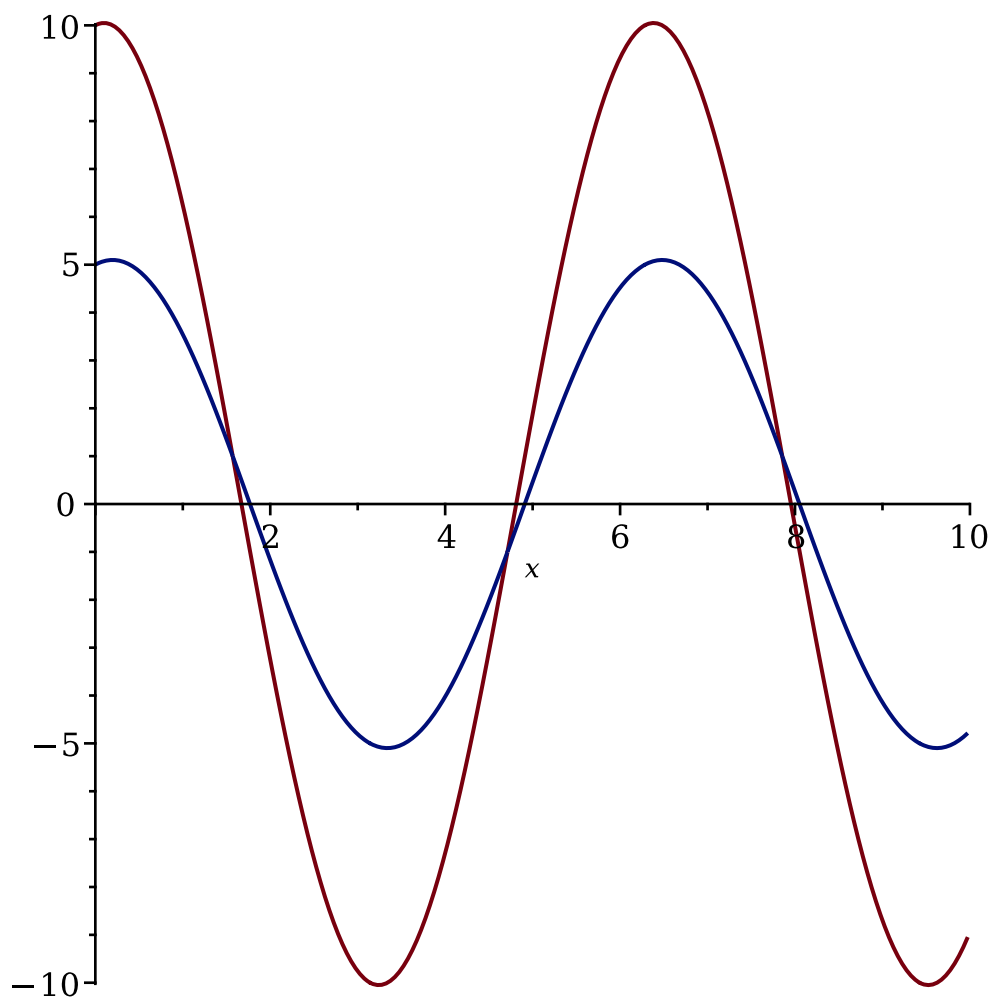


```

> with(DEtools):
> ecdif:=diff(y(x), x) + y(x)·tan(x) = 1/cos(x) :
> dsolve(ecdif, y(x));
      y(x) = (tan(x) + c1) cos(x)
> y1 := (x, c)→(tan(x) + c)·cos(x) :
> with(plots):
> plot([y1(x, 10), y1(x, 5)], x = 0..10)

```

**(6)**



```
> with(DEtools):
```

```
> ecdif := diff(y(x), x) = - \left( \frac{(x + y(x))}{y(x)} \right):
```

```
> dsolve(ecdif, y(x));
```

$$y(x) = \frac{\sqrt{3} x \tan\left(\text{RootOf}\left(\sqrt{3} \ln\left(\frac{3x^2}{4} + \frac{3x^2 \tan(Z)^2}{4}\right) + 2\sqrt{3} c_1 - 2Z\right)\right)}{2} - \frac{x}{2} \quad (7)$$

```
> soll := dsolve(ecdif, y(x));
```

$$\text{soll} := y(x) = \frac{\sqrt{3} x \tan\left(\text{RootOf}\left(\sqrt{3} \ln\left(\frac{3x^2}{4} + \frac{3x^2 \tan(Z)^2}{4}\right) + 2\sqrt{3} c_1 - 2Z\right)\right)}{2} - \frac{x}{2} \quad (8)$$

```
> right_hand_expr := rhs(soll);
```

$$\text{right\_hand\_expr} := \quad (9)$$

$$\frac{\sqrt{3} x \tan\left(\text{RootOf}\left(\sqrt{3} \ln\left(\frac{3 x^2}{4} + \frac{3 x^2 \tan(Z)^2}{4}\right) + 2 \sqrt{3} c_1 - 2 Z\right)\right)}{2} - \frac{x}{2}$$

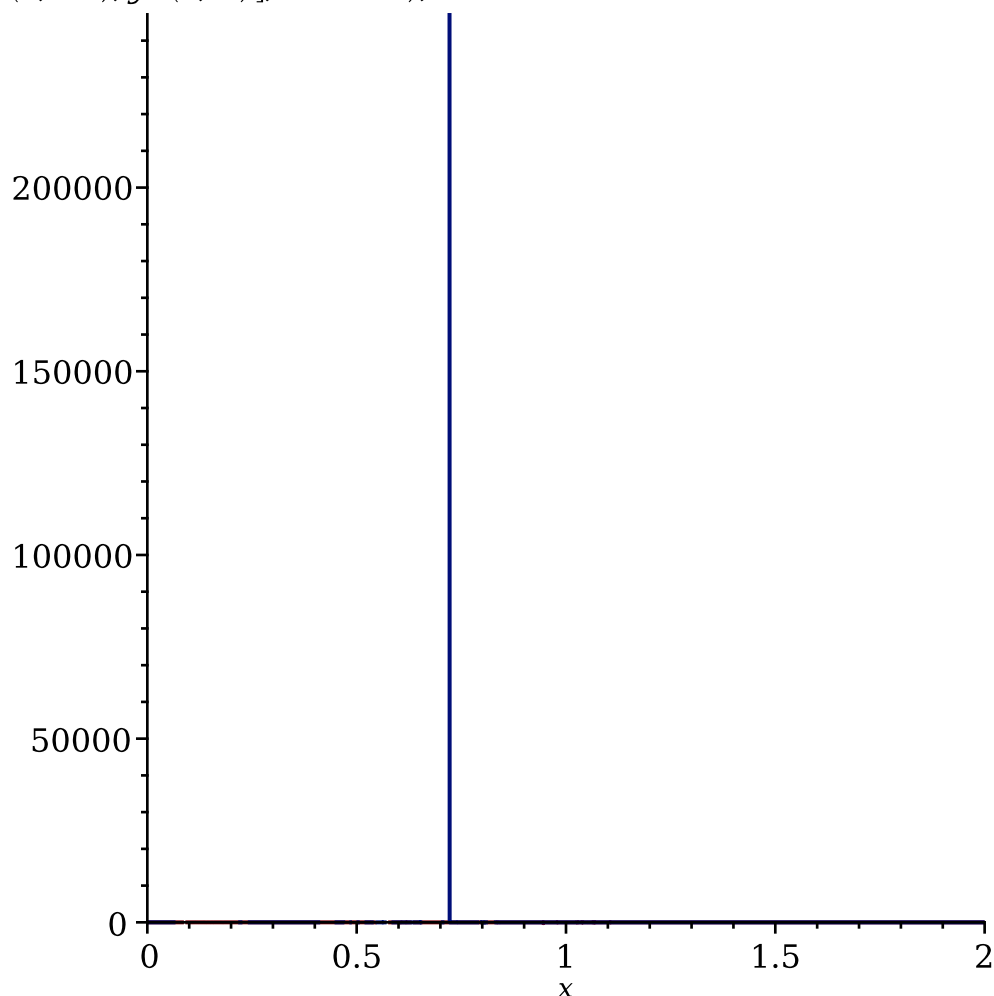
```
> y1 := unapply(right_hand_expr, x, _C1);
```

```
y1 := (x, c1)
```

(10)

$$\mapsto \frac{\sqrt{3} \cdot x \cdot \tan\left(\text{RootOf}\left(\sqrt{3} \cdot \ln\left(\frac{3 \cdot x^2}{4} + \frac{3 \cdot x^2 \cdot \tan(Z)^2}{4}\right) + 2 \cdot \sqrt{3} \cdot c_1 - 2 \cdot Z\right)\right)}{2} - \frac{x}{2}$$

```
> plot([y1(x, 10), y1(x, 3)], x = 0..2);
```



```
> with(DEtools) :
```

```
> ecdf := diff(y(x), x) + \frac{2}{x} \cdot y(x) = x^3 :
```

```
> dsolve(ecdf, y(x));
```

$$y(x) = \frac{\frac{x^6}{6} + c_1}{x^2}$$

(11)

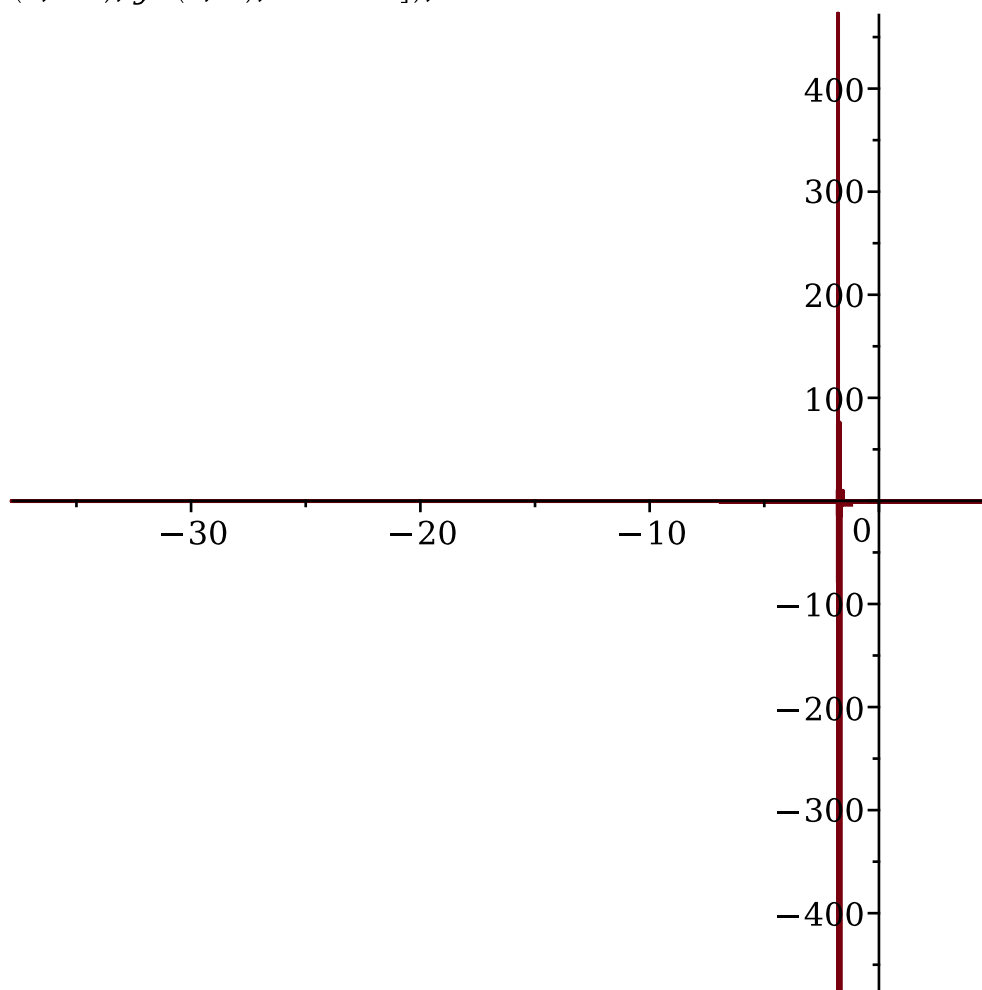
```
> y1 := (x, c) →  $\frac{\left(\frac{x^6}{6} + c\right)}{x^2} :$ 
```

$$y := (x, c) \mapsto \frac{\frac{x^6}{6} + c}{x^2}$$

**(12)**

```
> with(plots) :
```

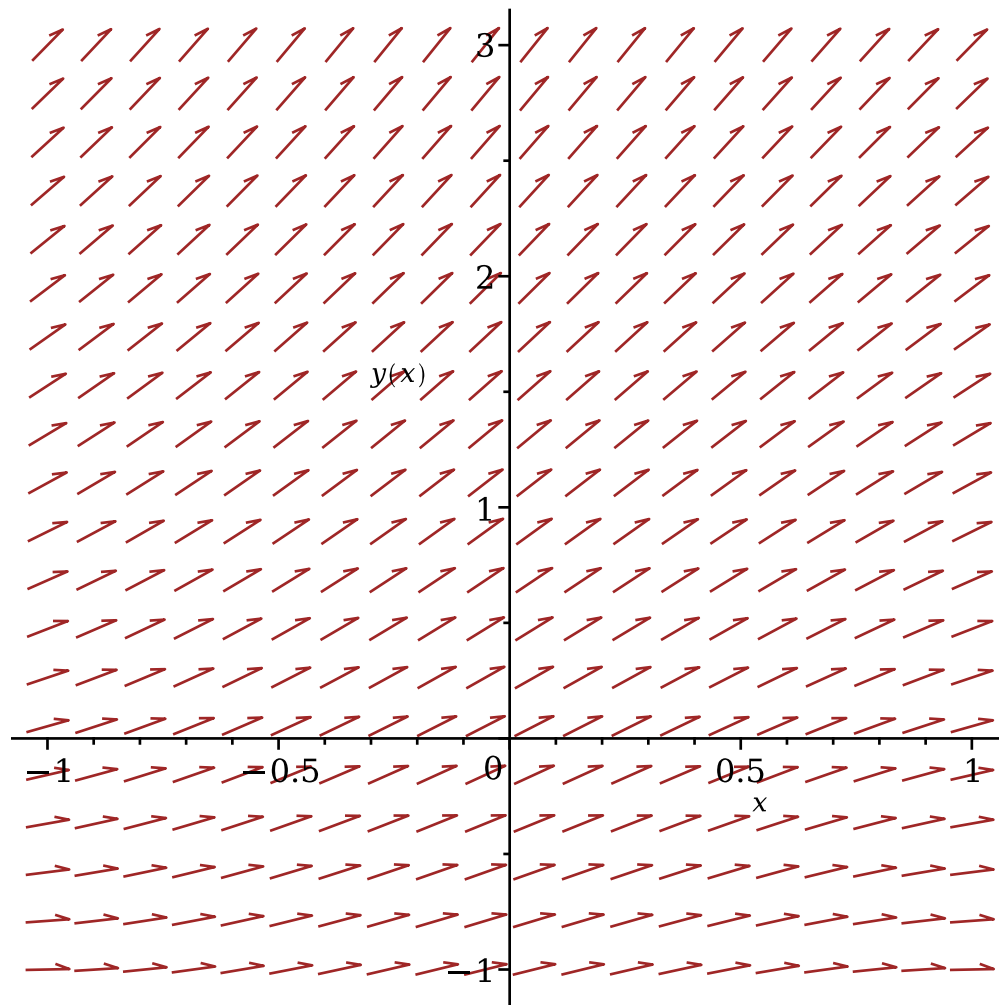
```
> plot([y1(x, 10), y1(x, 2), x = 1..3]);
```



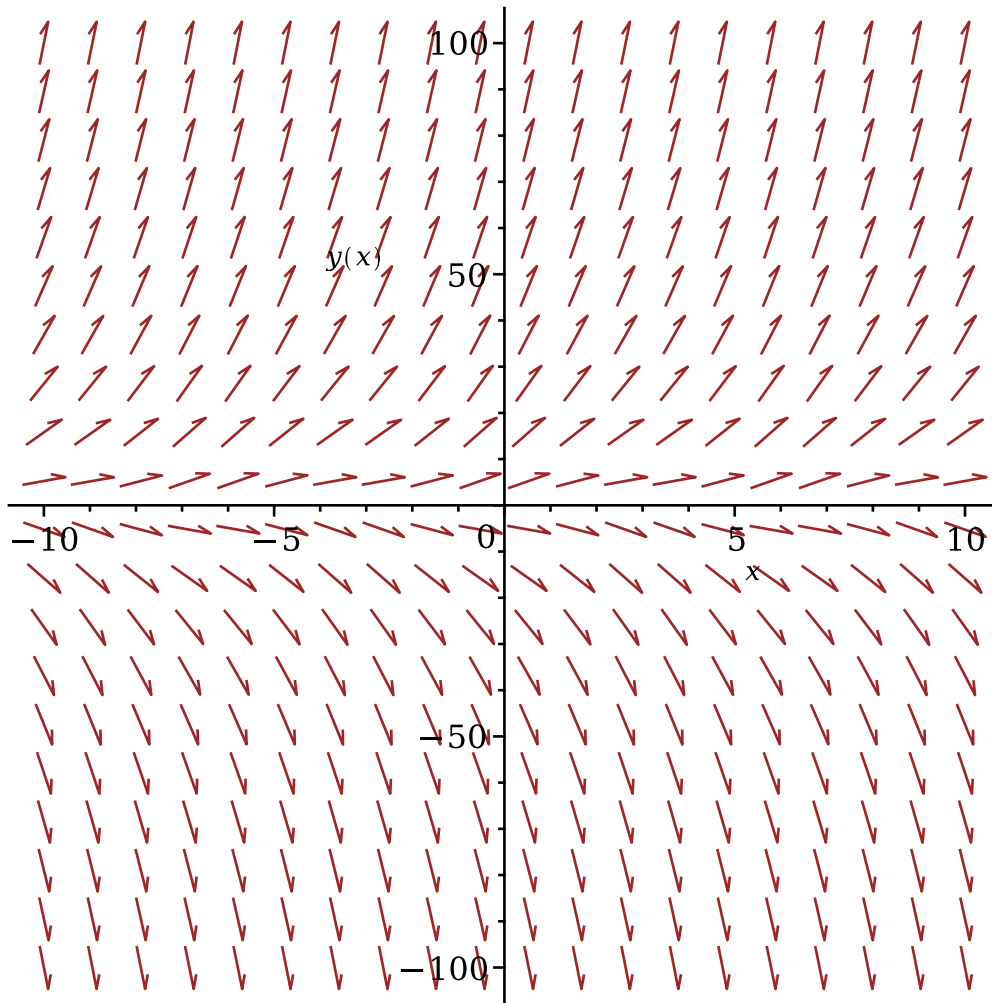
```
> with(DETools) :
```

```
> ecdif := diff(y(x), x) -  $\frac{1}{2} \cdot (y(x)) = \cos(x) :$ 
```

```
> DEplot(ecdif, y(x), x = -1..1, y = -1..3, stepsize = 0.1);
```



**>** `DEplot(ecdif, y(x), x = -10..10, y = -100..100, stepsize = 0.1)`



```
> soll := dsolve(ecdif, y(x));
```

$$soll := y(x) = -\frac{2 \cos(x)}{5} + \frac{4 \sin(x)}{5} + e^{\frac{x}{2}} c_1 \quad (13)$$

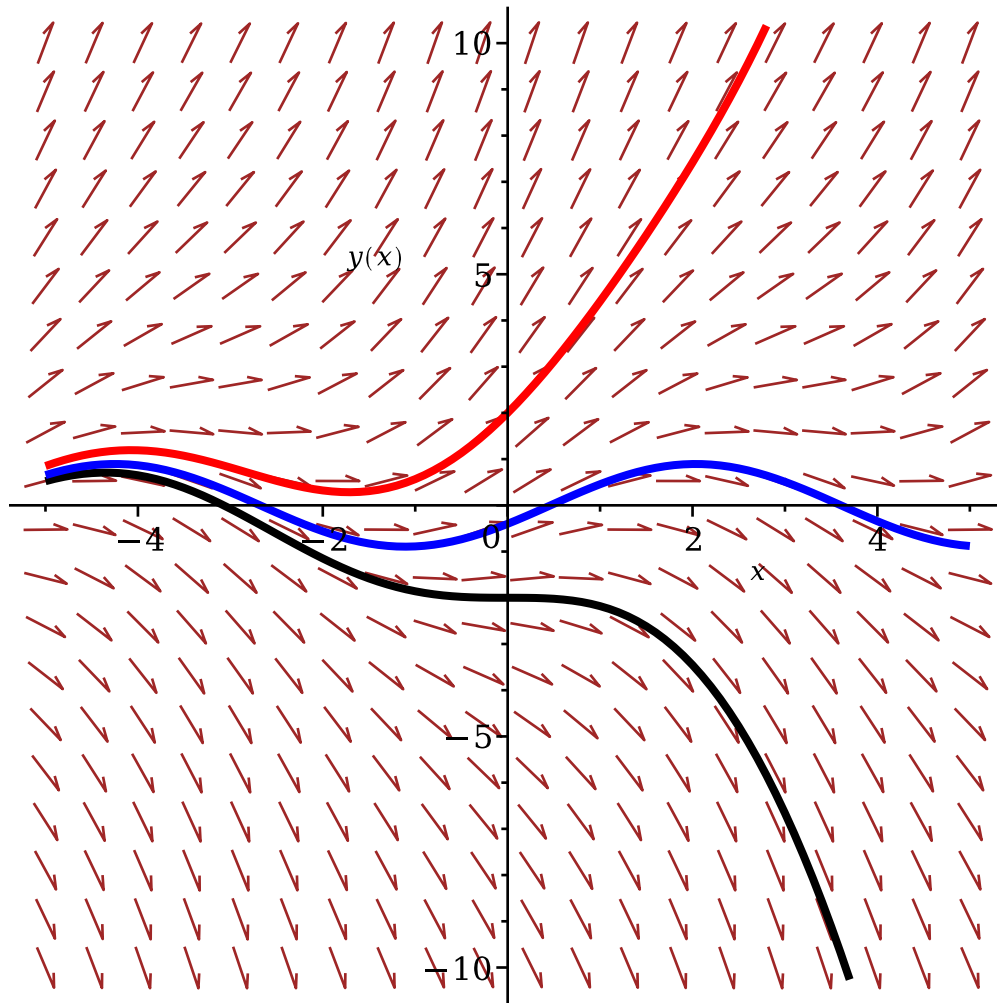
```
> right_hand_side := rhs(soll);
```

$$right\_hand\_side := -\frac{2 \cos(x)}{5} + \frac{4 \sin(x)}{5} + e^{\frac{x}{2}} c_1 \quad (14)$$

```
> y1 := unapply(right_hand_side, x, c)
```

$$y1 := (x, c) \mapsto -\frac{2 \cdot \cos(x)}{5} + \frac{4 \cdot \sin(x)}{5} + e^{\frac{x}{2}} \cdot c_1 \quad (15)$$

```
> DEplot(ecdif, y(x), x = -5..5, y = -10..10, [[y(0) = -2], [y(0) = 2], [y(0) = -2/5]],
         stepsize = 0.1, linecolor = [black, red, blue]);
```



```
> with(DETools) :
```

```
> ecdif := diff(y(x), x, x) + y(x) = sin(x) + cos(x);
```

$$ecdif := \frac{d^2}{dx^2} y(x) + y(x) = \sin(x) + \cos(x) \quad (16)$$

```
> soll := dsolve(ecdif, y(x));
```

$$soll := y(x) = \sin(x) c_2 + \cos(x) c_1 + \frac{(1-x) \cos(x)}{2} + \frac{\sin(x) x}{2} \quad (17)$$

```
> right_hand_side := rhs(soll)
```

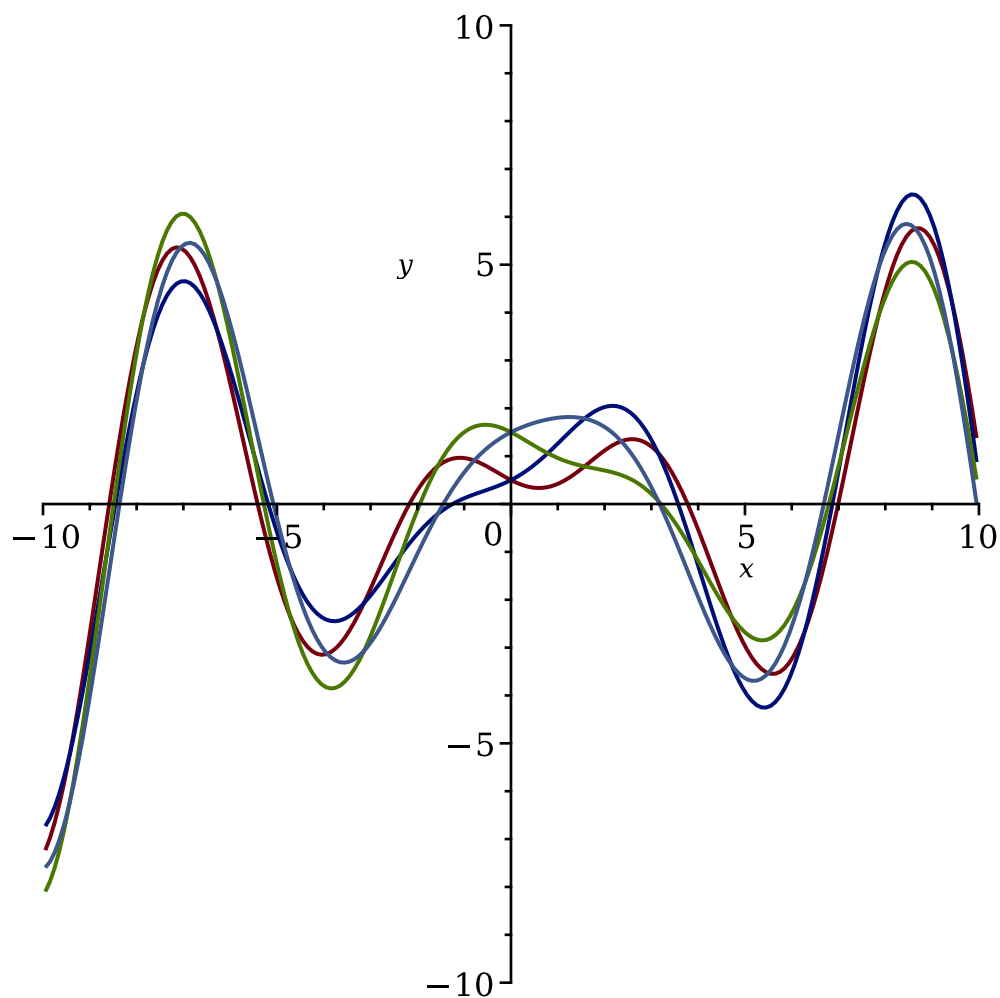
$$right\_hand\_side := \sin(x) c_2 + \cos(x) c_1 + \frac{(1-x) \cos(x)}{2} + \frac{\sin(x) x}{2} \quad (18)$$

```
> y1 := unapply(right_hand_side, x, c1, c2);
```

$$y1 := (x, c_1, c_2) \mapsto \sin(x) \cdot c_2 + \cos(x) \cdot c_1 + \frac{(1-x) \cdot \cos(x)}{2} + \frac{\sin(x) \cdot x}{2} \quad (19)$$

```
> plot([y1(x, 0, 0), y1(x, 0, 1), y1(x, 1, 0), y1(x, 1, 1)], x = -10..10, y = -10..10);
```





```
> with(DETools):
> ecdif := diff(y(x), x, x) - 5*diff(y(x), x) + 4*y(x) = 0;
      ecdif :=  $\frac{d^2}{dx^2} y(x) - 5 \frac{d}{dx} y(x) + 4 y(x) = 0$  (20)
```

```
> soll := dsolve(ecdif, y(x));
      soll :=  $y(x) = c_1 e^x + c_2 e^{4x}$  (21)
```

```
> right_hand_side := rhs(soll)
      right_hand_side :=  $c_1 e^x + c_2 e^{4x}$  (22)
```

```
> y1 := unapply(right_hand_side, x, c1, c2);
      y1 :=  $(x, c_1, c_2) \mapsto c_1 \cdot e^x + c_2 \cdot e^{4x}$  (23)
```

```
> cond_in := y(0) = 5, D(y)(0) = 8;
      cond_in :=  $y(0) = 5, D(y)(0) = 8$  (24)
```

```
> soll_1 := dsolve({ecdif, cond_in}, y(x));
      soll_1 :=  $y(x) = 4 e^x + e^{4x}$  (25)
```

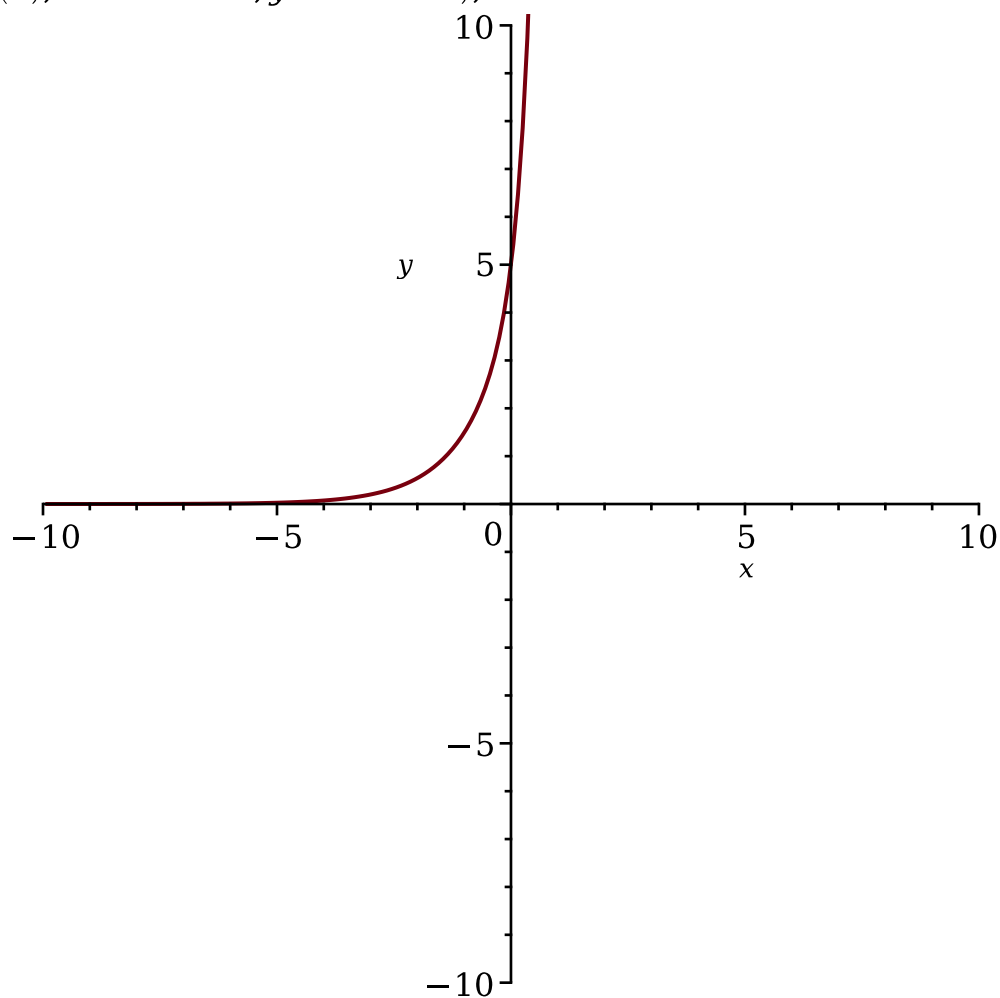
```
> right_hand_side_1 := rhs(soll_1):
```

```
> y2 := unapply(right_hand_side_1, x);
```

$$y2 := x \mapsto 4 \cdot e^x + e^{4 \cdot x}$$

(26)

```
> plot(y2(x), x = -10..10, y = -10..10);
```



```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x, x) - diff(y(x), x) - 2 * y(x) = 0:
```

```
> cond_in := y(0) = a, D(y)(0) = 2:
```

```
> soll := dsolve({ecdif, cond_in}, y(x));
```

$$soll := y(x) = \frac{(-2 + 2a)e^{-x}}{3} + \frac{e^{2x}(a+2)}{3}$$

(27)

```
> limit(soll, x = infinity)
```

$$\lim_{x \rightarrow \infty} y(x) = \text{signum}(a+2) \cdot \infty$$

(28)

```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x) + \frac{k}{x} \cdot y(x) = x^3:
```

```
> soll := dsolve(ecdif, y(x));
```

$$soll := y(x) = \frac{x^4}{4+k} + x^{-k} c_1$$

(29)

```
> right_hand_side := rhs(soll) :
```

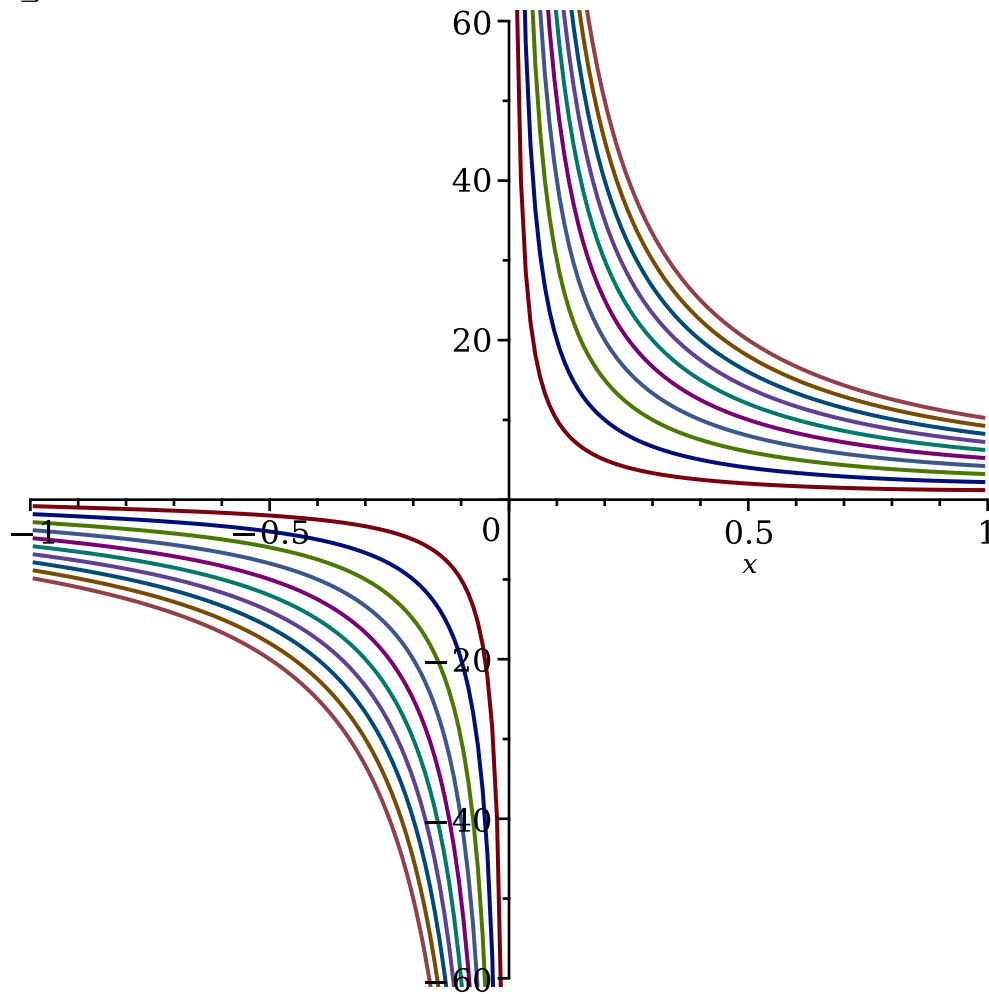
```
> y1 := unapply(right_hand_side, x, k, c1);
```

$$y1 := (x, k, c_1) \mapsto \frac{x^4}{4+k} + x^{-k} \cdot c_1 \quad (30)$$

```
> list_f := y1(x, 1, i) $ i = 1..10;
```

$$\text{list\_f} := \frac{x^4}{5} + \frac{1}{x}, \frac{x^4}{5} + \frac{2}{x}, \frac{x^4}{5} + \frac{3}{x}, \frac{x^4}{5} + \frac{4}{x}, \frac{x^4}{5} + \frac{5}{x}, \frac{x^4}{5} + \frac{6}{x}, \frac{x^4}{5} + \frac{7}{x}, \frac{x^4}{5} + \frac{8}{x}, \frac{x^4}{5} + \frac{9}{x}, \frac{x^4}{5} + \frac{10}{x} \quad (31)$$

```
> plot([list_f], x = -1..1);
```



```
> ecdif2 := diff(y(x), x) + 1/x * y(x) = x^3 :
```

Error. (in DEtools/DEplot) system must have same number of dependent variables as DE's.

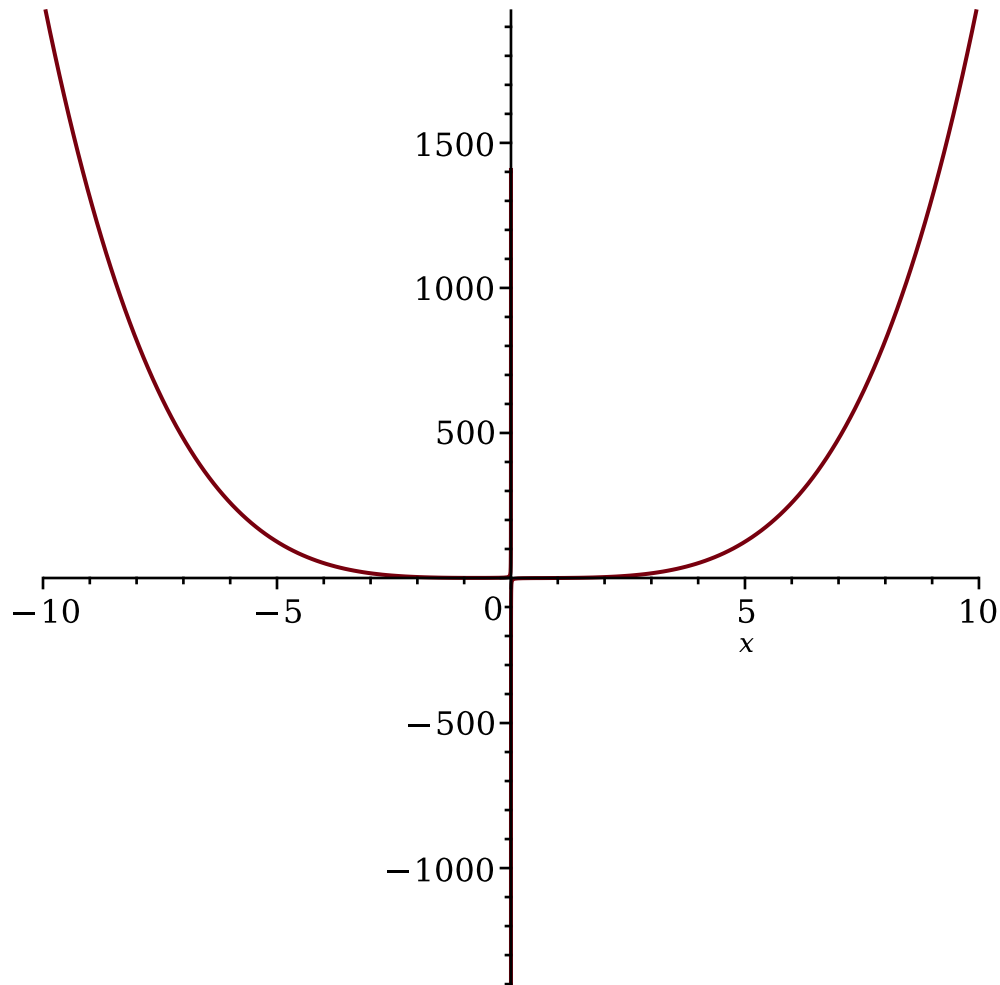
```
> soll_2 := dsolve(ecdif2)
```

$$\text{soll\_2} := y(x) = \frac{\frac{x^5}{5} + c_1}{x} \quad (32)$$

```
> DEplot(ecdif2, y(x), x = -10..10, y = -10..10)
DEplot( $\frac{d}{dx} y(x) + \frac{y(x)}{x} = x^3$ , y(x), x = -10..10, y = -10..10) (33)
```

```
> cond_in := y(1) = 0:
> soll_c := dsolve({ecdif2, cond_in}, y(x)):
> right_hand_side_c := rhs(soll_c):
> y_c := unapply(right_hand_side_c, x)
y_c := x ↦  $\frac{x^5 - 1}{5 \cdot x}$  (34)
```

```
> plot(y_c(x), x = -10..10)
```



```
> ecdif2 := diff(y(x), x) +  $\frac{k}{x} \cdot y(x) = x^3$ :
> soll_d := dsolve({ecdif2, cond_in}, y(x));
soll_d := y(x) =  $\frac{x^4 - x^{-k}}{4 + k}$  (35)
```

```
> right_hand_side_d := rhs(soll_d):
> y_d := unapply(right_hand_side_d, x, k);
y_d := (x, k) ↦  $\frac{x^4 - x^{-k}}{4 + k}$  (36)
```

```

> list_d := y_d(x, i)$i = 1..10;
list_d :=  $\frac{x^4}{5} - \frac{1}{5x}, \frac{x^4}{6} - \frac{1}{6x^2}, \frac{x^4}{7} - \frac{1}{7x^3}, \frac{x^4}{8} - \frac{1}{8x^4}, \frac{x^4}{9} - \frac{1}{9x^5}, \frac{x^4}{10}$ 
 $-\frac{1}{10x^6}, \frac{x^4}{11} - \frac{1}{11x^7}, \frac{x^4}{12} - \frac{1}{12x^8}, \frac{x^4}{13} - \frac{1}{13x^9}, \frac{x^4}{14} - \frac{1}{14x^{10}}$ 

```

(37)